| Register name | Abbreviation | Register's function |
|---|---|---|
| Current instruction register | CIR | Stores the current instruction while it is being decoded and executed |
| Index register | IX | Stores a value; only used for indexed addressing |
| Memory address register | MAR | Stores the address of a memory location or an I/O component which is about to have a value read from or written to |
| Memory data register (memory buffer register) | MDR (MBR) | Stores data that has just been read from memory or is just about to be written to memory |
| Program counter | PC | Stores the address of where the next instruction is to be read from |
| Status register | SR | Contains bits that are either set or cleared which can be referenced individually |

Table 5.01 Registers in a simple CPU

There are three important points to remember. The first is that the MDR must act as a buffer. This is because transfers of data inside the processor take place much more quickly than transfers outside the processor. The second point to note is that the index register (IX) can be abbreviated as IR but in some sources the current instruction register (CIR) is abbreviated as 'IR'. This is a potential cause of confusion. In this book, the index register is always IX and the current instruction register is CIR. Finally, there is also possible confusion if the abbreviation PC is used. This will only be used in this book when register transfer notation is being used, as you will see later in the chapter. Everywhere else, a PC is a computer.



For all of the special-purpose registers, except for the status register, the contents represent one value. For the status register each individual bit is used as a logical flag. The bit is set to 1 if a condition is detected. Examples are the carry flag, the negative flag and the overflow flag.

Chapter 6 (Section 6.07) contains some examples of the use of the accumulator and the status register.

## 5.03 The system bus

A bus is a parallel transmission component with each separate wire carrying a single bit. It is important not to describe a bus as a storage device. A bus does not hold data. Instead it is a mechanism for data to be transferred from one system component to another.

There will be buses inside the CPU. These are not considered here. The system bus connects the CPU to the memory and to the I/O system. In the simple computer system described in this chapter there will be a system bus that comprises three distinct components: the address bus, the data bus and the control bus. The schematic diagram of the CPU in Figure 5.01 shows the logical connection between each bus and a CPU component. The address bus is connected to the MAR; the data bus to the MDR; and the control bus to the control unit. The system bus allows data flow between the CPU, the memory and input or output (I/O) devices as shown in the schematic diagram in Figure 5.02.

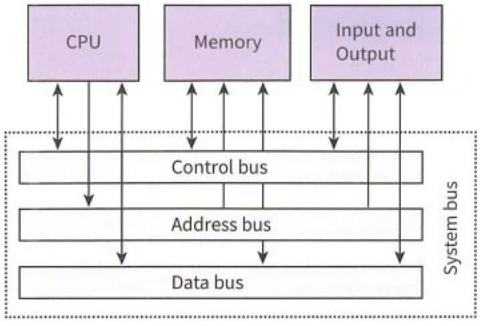



Figure 5.02 A schematic diagram of the system bus

## The address bus

The sole function of the **address bus** is to carry an address. This address is loaded on to the bus from the MAR as and when directed by the control unit. The address specifies a location in memory or an I/O component which is due to receive data or from which data is to be read. The address bus is a 'one-way street'. It can only be used to send an address to a memory controller or an I/O controller. It cannot be used to carry an address back to the CPU.

KEY TERM

**Address bus:** a component that carries an address. This can be to the memory controller to identify a location in memory which is to be read from or written to or it can be to the I/O system to identify the source or destination of the data



## The data bus

The function of the **data bus** is to carry data. This might be an instruction, an address or a value. As can be seen from Figure 5.02, the data bus is two-way (bidirectional): it might be carrying data from the CPU to the memory or carrying data to the CPU. However, another option is to carry data to or from an I/O device. The diagram does not make clear whether, for instance, data coming from an input device is carried first to the CPU or directly to the memory. There is a good reason for this. Some computer systems will only allow input to the CPU before the data can be stored in memory. Other systems will allow direct transfer to memory.

KEY TERM

**Data bus:** a component that can carry data from the processor to the memory or to an output device or can carry data from the memory or from an input device

## The control bus

The control bus is another bidirectional bus which transmits a signal from the control unit to any other system component or transmits a signal to the control unit. There is no need for extended width, so the control bus typically has just eight wires. A major use of the control bus is to carry timing signals. As described in Section 5.02, the system clock in the control unit defines the clock cycle for the computer system. The control bus carries timing signals at time intervals dictated by the clock cycle. This ensures that the time that one component transmits data is synchronised with the time that another component reads it.

## 5.04 Factors contributing to system performance

The processor clock speed could is a very important factor governing the processing speed of the system. This is because one clock cycle defines the shortest possible time that any action can take. Actually, none of the components outside of the processor can work anywhere near as fast as the processor can. The components that are directly addressable by the processor, which can be referred to as the immediate access store (IAS), can only accept data from or provide data to the processor at speeds much slower than the processor speed.

Because of this problem modern processors are far more complex than the simple example that has been discussed in this chapter. One example of this complexity is that the CPU chip or integrated circuit will be multi-core. Each core is a separate processor. Performance improves with increasing number of cores. A further factor is the use of cache memory which was briefly discussed in Chapter 3 (Sections 3.01 and 3.03). Cache memory is the fastest component of the IAS. Performance improves with increased storage size for the cache and with increased rate of access. Fastest access is obtained by having all or part of the cache on the CPU chip.

Before considering other factors, it is useful to introduce the concept of a **word**. A word consists of a number of bytes and for any system the word length is defined. The significance of the word length is that it defines a grouping that the system can handle as one unit. The word length might be stated as a number of bytes or as a number of bits. Typical word lengths are 16, 32 or 64 bits; that is, 2, 4 or 8 bytes, respectively. The word length will influence the system architecture design in regard to the capacity of the components. For example, it is usual for the size of registers to match the word length. Word length also has to be considered when making decisions about bus widths.





KEY TERM

**Word:** a small number of bytes that can be handled as a unit by the computer system

For the address bus, the bus width defines the number of bits in the address's binary code. In a very simple computer system the bus width might be 16 bits, allowing 65 536 memory locations to be directly addressed. Such a memory size would, of course, be totally inadequate for a modern computer system. Even doubling the address bus width to 32 bits would only allow direct addressing of a little over four billion addresses. As a result, special techniques are used when the storage capacity of the memory is too large for direct addressing. Their use affects system performance.

Bus width is again an important factor in considering how the data bus is used. For a given computer system, the data bus width is ideally the same as the word length. If this is not possible, the bus width can be half the word length so that a full word can be transmitted by two consecutive data transfers. Clearly the performance of the system is affected if the latter case applies.

### Extension Question 5.01

In an advertisement for a laptop computer, the system is described as 4 GB, 1 TB, 1.7 GHz.

**a** Which three components are being referred to here?

**b** Calculate the minimum time period that could separate successive activities on this system.

### Extension Question 5.02

Can you find out the bus widths used in the computer system you are using?

## 5.05 I/O ports

The schematic diagram in Figure 5.02 slightly misrepresents the system architecture because it looks as if the CPU, the memory and the I/O devices have similar access to the data and control buses. The reality is different. Each I/O device is connected to an interface called a port. Each port is connected to the I/O or device controller. This controller handles the interaction between the CPU and an I/O device. A port is described as 'internal' if the connected I/O device is an integral part of the computer system. An external port allows the computer user to connect a peripheral I/O device.

### The Universal Serial Bus (USB)

In the early days of the PC, the process of connecting a peripheral was not something the ordinary user would try to do; it required technical expertise. The aim of the plug-and-play concept was to remove the need for technical knowledge so that any computer user could connect a peripheral and start using it straight away. The plug-and-play concept was only fully realised by the creation of the Universal Serial Bus (USB) standard. Nowadays anyone buying a new peripheral device will expect it to connect to a USB port. There is an alternative technology known as FireWire, but this is not so commonly used in computer systems.



**TIP**

Don't forget that the USB is a bus. A USB drive stores data and is connected to a USB port which allows data to be transmitted along the bus.



The following is some information about the USB standard.

- A hierarchy of connections is supported.
- The computer is at the root of this hierarchy and can handle 127 attached devices.
- Devices can be attached while the computer is switched on and are automatically configured for use.
- The standard has evolved, with USB 3.2 being the latest version.

**Discussion Point:**

Carry out an investigation into storage devices that could be connected as a peripheral to a PC using the USB port.

For two representative devices find out which specific USB technology is being used and what the potential data transfer speed is. How do these speeds compare with the speed of access of a hard drive installed inside the computer?

### Specialised multimedia ports

Despite the widespread use of USB ports there are some peripheral devices that require a different port, one that is specialised for the type of device. Although computer systems come packaged with a monitor for screen display there is sometimes a requirement for a second screen to be used. The connection of the second screen can be through a Video Graphics Array (VGA) port. This provides high-resolution screen display which is suitable for most display requirements. However, if the screen is needed to display a video, the VGA port is not suitable because it does not transmit the audio component.

A High Definition Multimedia Interface (HDMI) port will provide a connection to a screen and allow the transmission of high-quality video including the audio component.